

ALL INDIA INSTITUTE OF MEDICAL SCIENCES
BHUBANESWAR, ODISHA
DEPARTMENT OF PHYSIOLOGY
ANNOUNCES



WORKSHOP 2025

The XXXVI Annual Conference of
The Physiological Society of India
"Innovations in Physiology: From Cell to Systems"



For More Information Visit : <https://physicon2025.com/>

For Registration Visit :
<https://register.physicon2025.com/>

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Facilitator: Dr Basanta Manjari Naik, Additional Professor, JIPMER, Puducherry

WORKSHOP - 1

MASTER THE DECODING SKELETAL MUSCLE HEALTH

Join Us for a Full day Workshop on
Skeletal Muscle Health

20

Thursday,
November 2025

08:30am - 05:30pm



KEY TOPIC COVERED:

- Skeletal Muscle in Metabolism
- Electromyography & Nerve conduction study
- DEXA & BIA
- Skeletal Muscle in Paediatric Age Group
- Human Gait and Role of Skeletal Muscles
- Skeletal Muscle Proteomics



HANDS ON SESSIONS

- Sarcopenia Status Determination
- Immunohistochemical Analysis of Skeletal Muscle



REGISTRATION FEE FOR WORKSHOP 1

INR 2000

FULL DAY WORK SHOP



Phone

+91-9938569067



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Venue

- Hematology Lab and Clinical Physiology Lab, Department of Physiology (Forenoon)
- IHC Lab , Department of Anatomy (afternoon)

PHYSICON 2025



DECODING SKELETAL MUSCLE HEALTH
CLINICAL INSIGHTS & IMMUNOHISTOCHEMISTRY
ANALYSIS OF AGE-RELATED FIBRE TYPE SHIFT

WORKSHOP



8:30 AM - 9 AM	REGISTRATION & BREAKFAST
9 AM - 9:15 AM	OPENING CEREMONY
SESSION I	CLINICAL TOOLS FOR ASSESSING MUSCLE HEALTH AND DISEASE
9:15 AM - 9:45 AM	Role of Skeletal Muscle in metabolism via Interorgan Crosstalk Dr Rashmiranjan Mohanty, Professor, Department of General Medicine, AIIMS, Bhubaneswar
9:45 AM - 10:45 AM	Hands on interactive session: Skeletal Muscle Evaluation and Sarcopenia Status Determination Dr Manas Kumar Panigrahi, Additional Professor and HOD, Department of Gastroenterology, AIIMS, Bhubaneswar Dr Nibedita Priyadarsini, Associate Professor, Department of Physiology, AIIMS, Bhubaneswar
10:45 AM - 11AM	Skeletal Muscle Assessment in the Paediatric Age Group Prof (Dr) Manaswini Mangaraj, Professor and HOD, Department of Biochemistry, AIIMS, Bhubaneswar
	TEA BREAK : 11:00 AM - 11:15 AM
11:15 AM - 11:45 AM	Hands-on session: Role of Electromyography and Nerve conduction study in assessing skeletal muscle function Dr B. N. Rao, Additional Professor, Department of Physiology, AIIMS, Bhubaneswar
11:45 AM - 12PM	Practical Exercise Session : Prevention of Age Related Muscle Loss Dr Sayantan Ray, Assistant Professor, Department of Endocrinology, AIIMS, Bhubaneswar
12 PM - 12:30 PM	Assessment Techniques for Bone and Muscle Metrics: Insights from DEXA and BIA: Visit to the Department of Endocrinology DEXA Lab Dr Kishore Kumar Behera, Additional Professor and HOD, Department of Endocrinology, AIIMS, Bhubaneswar
12:30 PM - 1 PM	Human Gait and Role of skeletal muscles in normal gait cycle: Visit to the PMR Department Gait lab Prof (Dr) Jagannath Sahoo, Professor and HOD, Department of PMR, AIIMS, Bhubaneswar
	LUNCH BREAK : 1 PM - 1:30 PM



DECODING SKELETAL MUSCLE HEALTH

CLINICAL INSIGHTS & IMMUNOHISTOCHEMISTRY
ANALYSIS OF AGE-RELATED FIBRE TYPE SHIFT

WORKSHOP

SESSION II

FIBER-TYPE DYNAMICS & IHC INSIGHTS IN SARCOPENIA

1:30 PM - 1:45 PM

Understanding muscle fiber-type dynamics during health and disease

[Dr Naresh Chandra Bal](#), Associate Professor,
Department of Biotechnology, KIIT University

1:45 PM - 2:15 PM

Skeletal muscle biopsy tissue handling (Immunohistochemistry)

- Unveiling Immunohistochemistry: Principles and Modern application [1:45 PM - 2 PM]

[Dr Ravi Kant Narayan](#), Assistant Professor,
Department of Anatomy, AIIMS, Bhubaneswar

- Immunohistochemistry protocol steps Demystified (Video demo) [2 PM - 2:15 PM]

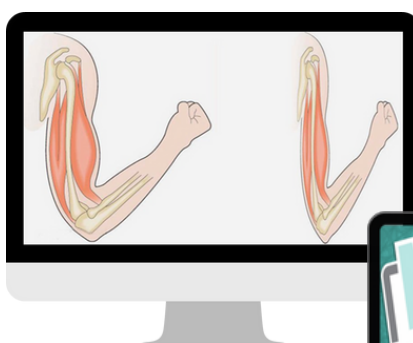
[Dr Praveen Kumar R](#), Assistant Professor,
Department of Anatomy, AIIMS, Bhubaneswar

2:15 PM - 5 PM

Hands-on session

5 PM - 5:30PM

Certificate distribution followed by tea



CME CUM WORKSHOP 2

(Full Day)

NOVEMBER 20, 2025 (THURSDAY)

9 AM - 5 PM

REGISTRATION FEE
INR 2000

Seminar Room & Room No. 38, Department
of Physiology, AIIMS, Bhubaneswar

CONVENOR
DR. KALPANA BARHWAL

physio_kalpana@aiimsbhubaneswar.edu.in

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WIRING THE BRAIN IN VITRO: ADVANCED TECHNIQUES FOR NEURONAL CULTURE AND PROTEIN EXPRESSION ANALYSIS

WORKSHOP

2

8:30 AM - 9 AM

REGISTRATION & BREAKFAST

SESSION I

ESTABLISHING PRIMARY NEURONAL NETWORKS – OPTIMIZED
PROTOCOLS FOR CULTURING FUNCTIONAL NEURONS IN VITRO
(FORENOON)

9 AM - 9:30 AM

‘Building Brains in a Dish: How In-Vitro Models Reveal
Neural Connectivity’

Seminar Room, Department
of Physiology, Ground Floor,
Academic Block

Dr. Kalpana Barhwal, Additional Professor
Department of Physiology, AIIMS Bhubaneswar

9:30 AM - 10 AM

Quantitative Proteomics Reveals Brain Protein
Alterations in Rabies Virus Infection

Dr Amol Suryavanshi, ILS Bhubaneswar

10 AM - 10:30 AM

Human neural stem cells and Cerebral Brain Organoids
as models to study virus induced neurodegeneration

Dr. Pankaj Seth, Scientist VII
(NBRC, Manesar, Gurugram)

10:30 AM - 11:15 PM

Hands-on session on :

- Establishing primary hippocampal neuronal culture
from newborn rat pups
- Immunocytochemistry

Room no.38 Department of
Physiology, Ground Floor,
Academic Block

Resource person: Mrs. Lakshmi Priya Panda (AIIMS Bhubaneswar)

TEA BREAK : 11:15 AM - 11:30 AM

11:30AM - 1 PM

Hands-on training on Quantification and Analysis of
Immunofluorescence images through Image J

Abhijit Bhattacharjee
Technical expert Olympus

LUNCH BREAK : 1 PM - 2 PM

SESSION II

QUANTITATIVE AND QUALITATIVE PROTEIN PROFILING (AFTERNOON)

2 PM - 4 PM

Blots and Bands: Demistifying Protein Quantification

Seminar Room, Department
of Physiology, Ground Floor,
Academic Block

Dr Swati Shrivastava,
Scientist E, DIPAS, DRDO, Delhi

4 PM - 5 PM

Beyond the Bands : Mastering Western Blot Detection
and Interpretation

Dr Surayanarayan Biswal, Assistant professor
Central University of Punjab, Bathinda
Certificate Distribution

5 PM - 5:30 PM

HIGH TEA: 5:30 ONWARDS

PHYSICON 2025

CME CUM WORKSHOP 3

(Forenoon)

NOVEMBER 20, 2025
(THURSDAY)

8:30 AM - 1:30 PM

REGISTRATION FEE
INR 1000

LT-2, Ground floor, Clinical Physiology
and UG Lab, Department of Physiology

CONVENOR
PROF.(DR.) JAYANTI MISHRA

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NEURODIAGNOSTICS IN ACTION: AUTONOMIC TESTING DEMYSTIFIED

WORKSHOP



8:30 AM - 9 AM	REGISTRATION & BREAKFAST
9 AM - 9:15 AM	INAUGURATION OF WORKSHOP AND GROUP PHOTO SESSION
SESSION I	LT-4, Ground Floor, Academic Block
9:15 AM - 10:00AM	Introduction to Resource Faculty members and Residents to the participants
10:00 AM - 10:30 AM	Understanding the basics of Heart Rate Variability (HRV) Dr. Manisha Kar, Professor and Head Department of Physiology, AIIMS Bhubaneswar
10:30 AM - 11AM	Introduction to Autonomic Function Tests Dr. Jayanti Mishra, Professor Department of Physiology, AIIMS Bhubaneswar
	TEA BREAK : 11:00 AM - 11:15 AM
11:15 AM - 12:45 AM	Demonstration and hands-on exposure to various techniques of Autonomic Function and HRV Dr. Aswin PR, Assistant Professor Department of Physiology SUT Academy of Medical Sciences, Trivandrum Dr. Mrudula (Junior Resident) Dr. Ajaykumar T (Junior Resident) Dr. Manthru (Junior Resident) Ms. Shubhlakshmi (Laboratory personnel)
	Clinical Physiology Laboratory, Department of Physiology, Ground Floor, Academic Block
12:45 PM - 1PM	Distribution of the certificates to the participants LT-4, Ground Floor, Academic Block
1 PM - 1:30 PM	LUNCH BREAK
	College of Nursing

CME CUM WORKSHOP 4 (Afternoon)

CONSTRUCTING INCLUSIVE
STUDENT-CENTERED LEARNING
SESSIONS (ISCLS) IN
PHYSIOLOGY: THROUGH THE
LENS OF UNIVERSAL DESIGN
AIDED BY AI TOOLS

NOVEMBER 20, 2025
(THURSDAY)

1 PM - 5 PM

REGISTRATION FEE
INR 500

Mini Auditorium,
AIIMS,Bhubaneswar

CONVENOR
DR. PRASUNPRIYA NAYAK,
PROFESSOR
nprasunpriya@gmail.com
+91-9963040449

CO-FACILITATOR
DR. SARMISHTHA GHOSH,
PROFESSOR (HEALTH
PROFESSION EDUCATION),
INS, BHAIKAKA UNIVERSITY,
KARAMSAD.

WORKSHOP

4

1 PM - 2 PM

REGISTRATION & LUNCH

2 PM - 2:30 PM

CLASSROOM SCENARIOS –
PARTICIPANT TO IDENTIFY AS
INCLUSIVE/EXCLUSIVE – JUSTIFY

Prof (Dr) Sarmishtha Ghosh

Mini Talk

Definition of ISCLS

Overview of Universal Design for Learning (UDL) principles

2:30 PM - 3 PM

Understanding Learner Diversity in the
Physiology Classroom

Prof (Dr) Sarmishtha Ghosh

Prof (Dr) Prasunpriya Nayak

3 PM - 3:25 PM

Universal Design Framework in Physiology Teaching

Prof (Dr) Prasunpriya Nayak

Prof (Dr) Sarmishtha Ghosh

TEA BREAK : 3:25 PM - 3:30 PM

3:30 PM - 4 PM

AI Tools for Inclusive Learning Design

Prof (Dr) Sarmishtha Ghosh

4 PM - 4:30 PM

Participants to try hands-on. Volunteers present
their AI-generated product.

Prof (Dr) Prasunpriya Nayak

4:30 PM- 5 PM

Wrap up / QA/ Feedback

Prof (Dr) Sarmishtha Ghosh

Dr Prasunpriya Nayak

CME CUM WORKSHOP 5

(Forenoon)

FROM STILLNESS TO
STRENGTH: HATHA YOGA,
HEARTFULNESS AND INNER
AWARENESS

NOVEMBER 20, 2025
(THURSDAY)

8:30 AM - 1 PM

REGISTRATION FEE
INR 500

Mini Auditorium,
AIIMS, Bhubaneswar

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DR. VARUN MALHOTRA
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WORKSHOP

5

8:30 AM - 9 AM

REGISTRATION & BREAKFAST

9 AM - 9:15 AM

OPENING CEREMONY

9:15 AM - 9:45 AM

Mapping the Mind-Body Constitution: Yogic
Assessment for Holistic

SESSION I

Dr Vandana S Daulatabad

9:45 AM - 10:45 AM

Guided Meditation session

SESSION II

Dr Varun Malhotra

10:45 AM - 11:30 AM

Heartfulness meditation

SESSION III

Dr Senthil Kumar Subramanian

TEA BREAK : 11:00 AM - 11:15 AM

11:30 AM - 12:30 PM

Stretch, Breathe and Be: Chair yoga for daily
balance

Suprabha Sharma

12:30 PM - 1 PM

Mapping the Mind-Body Constitution: Yogic
Assessment for Holistic Wellness

Dr Vandana Daulatabad

Group Discussion

LUNCH BREAK : 1 PM - 1:30 PM

PHYSICON 2025

CME CUM WORKSHOP 6

(Afternoon)

**POLYSOMNOGRAPHY A GOLD
STANDARD DIAGNOSTIC TOOL
FOR SLEEP STUDIES**

**NOVEMBER 20, 2025
(THURSDAY)**

1 PM - 5:30 PM

**REGISTRATION FEE
INR 1000**

**Haematology Lab,
Department of Physiology**

CONVENOR
DR BASANTA MANJARI NAIK
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+91-9488083671

WORKSHOP



1 PM - 2 PM

REGISTRATION & LUNCH

2 PM - 2:30 PM

Fundamentals of Sleep and Introduction to Polysomnography

Overview of sleep physiology and stages Classification of sleep disorders (ICSD-3) Historical background and evolution of PSG

Dr Trina Sengupta
Assistant Professor, ESIC Joka

2:30 PM - 3 PM

Indications, Preparation, and Protocols for PSG

Clinical indications for PSG (OSA, insomnia, parasomnias, etc.) Patient preparation and pre-study guidelines Components measured in PSG: EEG, EOG, EMG, ECG, respiratory effort, oxygen saturation Technical setup and sensor placement

Dr Archana Gaur T
Associate Professor, AIIMS Bibinagar

TEA BREAK : 3 PM- 3:15 PM

3:15 PM - 4:15 PM

Interpretation of PSG I

Scoring and staging based on AASM guidelines Reading and scoring a PSG report Identifying abnormalities in sleep architecture

Dr Basanta Manjari Naik
Additional Professor and HoD, JIPMER, Puducherry

4:15 PM - 5:15 PM

Interpretation of PSG II and case discussion

Diagnostic criteria for OSA, PLMS, RBD, narcolepsy, etc. Case-based examples and common pitfalls Home sleep apnea testing (HSAT) vs in-lab PSG Role of AI, wearable technology, and telemonitoring

Dr Senthil Velou
Additional Professor, AIIMS Madurai

5:15 PM - 5:30 PM

Question & Answer session

HIGH TEA : 5:30 PM ONWARDS